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Limitations

The biggest three limitations that I can think of using anonymous self-reported data are:

1. Verification
2. Selection bias
3. Nonresponse

The verification problem is quite straightforward, given that it is anonymous and self-reported, there are self-aggrandizing entries and/or unserious entries that are just inherent to the dataset. There is no effective way to ensure that the entries are true. The second problem is selection bias, gradcafe may attracts a certain kind of people on the internet, for instance, they could be more vested into the grad admissions process, or that there have more exposure to tech fields, or simply that they could be more among those who are high achievers—thus these populations are going to be overrepresented for any kind of statistics involving grad admissions, simply put, the IID assumption needed for sampling statistics is violated. Lastly, the problem of nonresponse— as this is a self-reported survey which is at the mercy of the respondent's patience and eagerness to fill out the fields, if the page takes too long to load, or if the respondent forgot their scores or term dates...etc they may get demotivated and put in short answers and do the bare minimum, leaving a lot of fields empty.

Let's look at the question, "what is the percentage of international students among those accepted at NYU's master program?" typical international student applying to NYU's master program may not likely enter their data onto gradcafe, perhaps they have not heard of it or perhaps they don't feel very confident in their scores and don't want to post their scores on the website. The website would probably encourage respondents who are more likely to be American, with stronger grades, and who got accepted to add into the entries. Mathematically, the average for the sample in gradcafe would likely skew up a fair bit with the distribution concentrating more heavily towards the higher ranges and likely with few or no representation for schools that are not well known. Nevertheless, the database can be used to elucidate many useful things about the real world such as interest in grad school, historic timelines for application decisions, and getting grade ranges for school acceptances (we assume higher score would always be better but seeing the range of low grades with acceptances can be insightful too if we assume people don't often report weaker/lower grade facetiously).